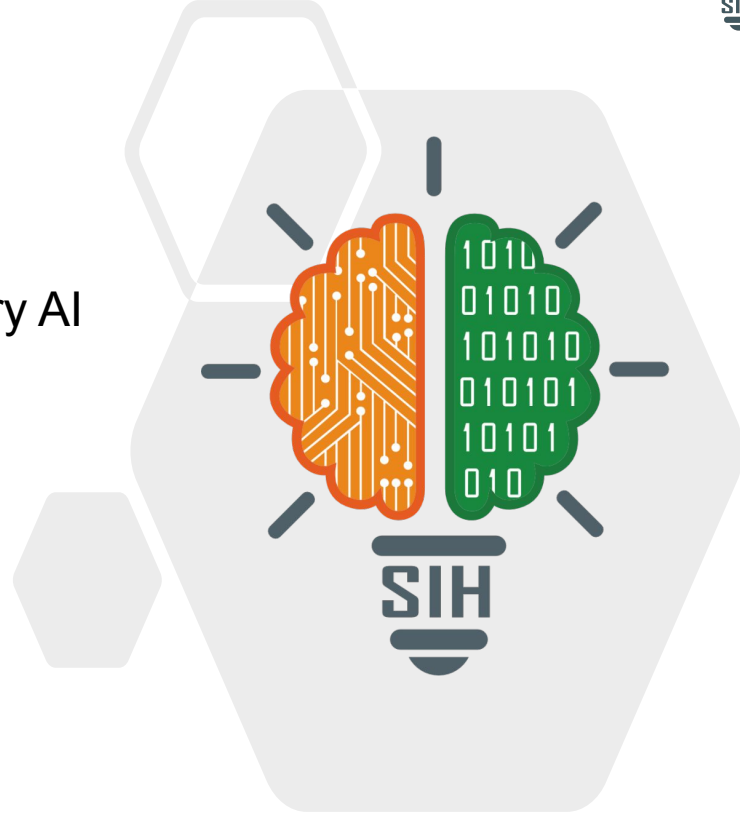


SMART INDIA HACKATHON 2026



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2026

- **Problem Statement ID** - SIH26167
- **Problem Statement Title** - SatQuery AI
- **Theme** - Space Technology
- **PS Category** - Software
- **Team Name** - Avengers

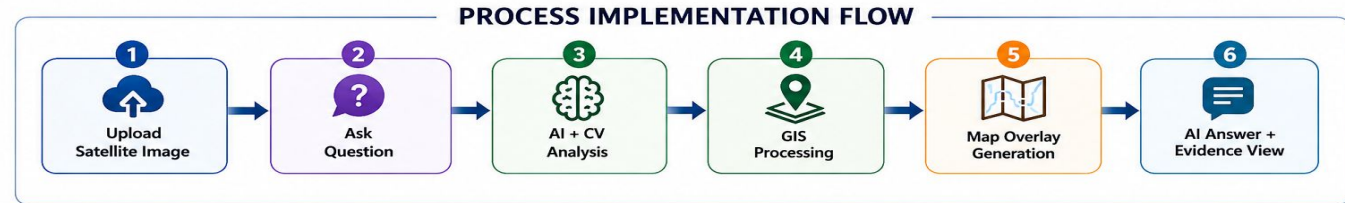
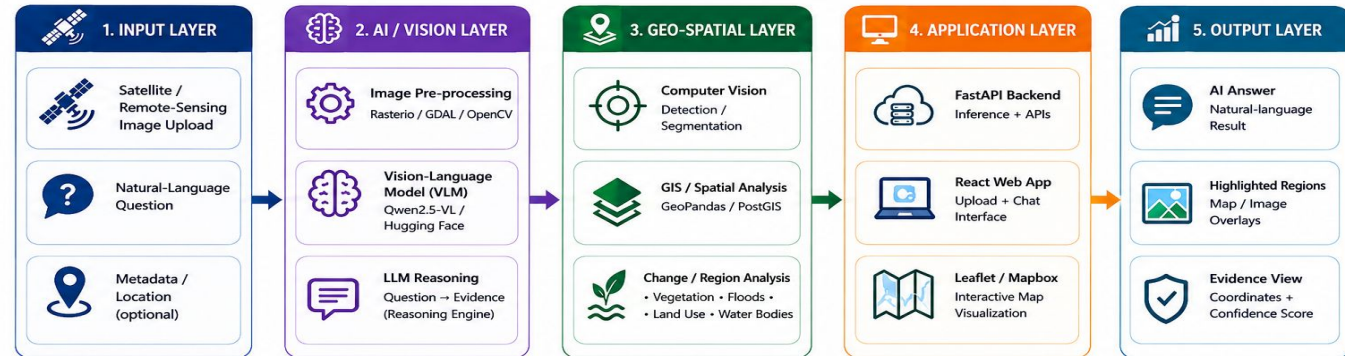


Proposed Solution

- Natural-language interface to query satellite and geospatial data, converting user queries into automated data selection and analysis workflows.
- Integrates satellite imagery, geospatial processing and AI/ML for analysis, delivering results through interactive maps, visualizations and concise insights.
- Eliminates the need for complex GIS/satellite-data expertise while automating data discovery, preprocessing and analysis.
- Reduces the time and effort required to extract insights, enabling researchers, officials and other users to access actionable satellite insights easily.
- Natural Language → Geospatial Analysis pipeline with AI-driven conversion of queries into relevant analytical workflows.
- Supports multi-source satellite/geospatial datasets and combines automated analysis, visual results and human-readable insights.

- **Frontend:** React.js + Leaflet/Mapbox
- **Backend:** Python + FastAPI
- **AI/ML:** Vision-Language Model (VLM) + LLM using PyTorch/Hugging Face
- **Image Processing:** OpenCV + Rasterio + GDAL
- **GIS/Spatial Analysis:** GeoPandas + PostGIS
- **Database:** PostgreSQL/PostGIS + Object Storage
- **Deployment:** Docker + Cloud

SYSTEM ARCHITECTURE – SATQUERY AI



Feasibility

- Open satellite datasets + proven AI/ML & GIS technologies
- Scalable, modular architecture
- Focused MVP for faster implementation

Challenges

- Large datasets → high processing requirements
- Ambiguous queries → inaccurate analysis
- Cloud cover & limited labelled data → reduced accuracy

Solutions

- Tiling, filtering & selective data retrieval
- Query validation + predefined workflows
- Cloud masking + transfer learning models



Impact

- **Researchers & Scientists:** Faster satellite insights without advanced GIS expertise.
- **Government & Disaster Management:** Faster detection of floods and land-use changes for timely response.
- **Decision-Makers:** AI-driven evidence for faster, informed decisions.

Benefits

- **Economic:** Reduces time, cost and technical effort in satellite-data analysis.
- **Social:** Makes complex geospatial insights accessible to non-experts.
- **Environmental:** Supports monitoring of deforestation, water bodies and vegetation changes.



- **RemoteCLIP** — Vision-language foundation model for remote-sensing image understanding
- **GeoChat** — Grounded vision-language model for remote-sensing image and spatial reasoning
- **Remote-Sensing Change Detection** — Research on temporal change analysis using satellite imagery
- **RSVQA** — Visual Question Answering for Remote Sensing Data — <https://arxiv.org/abs/2003.07333>
- **Copernicus Sentinel Data Space** — Sentinel-1 & Sentinel-2 satellite data — <https://dataspace.copernicus.eu/>